Project Name: Smart Media Management System

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List of Entities and their attributes :

Entity 1 — USERS

id (PK)
name
email (UNIQUE)
password_hash
role
created_at
updated_at
status
phone
username
avatar_color
avatar_url
semester
email_verified

Entity 2 — BOOKS

id (PK)
title
author
description
cover_url
pdf_url
category_id (FK → CATEGORIES.id)
preview_pages
total_pages
rating
language
publish_year
tags
status
created_at
updated_at
created_by (FK → USERS.id)
pdf_hosted_path
is_restricted

Entity 3 — VIDEOS

id (PK)
title
description
type
category_id (FK → CATEGORIES.id)
subcategory_id
thumbnail_url
tags
source_type
source_resource_key
status
access_level
view_count
release_date
created_by (FK → USERS.id)
created_at
updated_at

Entity 4 — WATCH_HISTORY

id (PK)
user_id (FK → USERS.id)
video_id (FK → VIDEOS.id)
progress_seconds
updated_at
UNIQUE(user_id, video_id)

Entity 5 — USER_CUSTOM_ROLES

user_id (PK, FK → USERS.id)
custom_role_id (PK, FK → CUSTOM_ROLES.id)
assigned_at

Entity 6 — SEARCH_HISTORY

id (PK)
user_id (FK → USERS.id)

query
content_type
created_at

Entity 7 — PYQ_PAPERS

id (PK)
semester
subject_name
course_code
exam_type
batch_info
file_urls
file_type
description
status
is_restricted
sort_order
created_by (FK → USERS.id)
created_at
updated_at
file_hosted_paths

Entity 8 — PLAYLISTS

id (PK)
title
description
thumbnail_url
created_by (FK → USERS.id)
is_public
source_type
source_url
created_at
updated_at
category_id (FK → CATEGORIES.id)

Entity 9 — PLAYLIST_ITEMS

id (PK)

playlist_id (FK → PLAYLISTS.id)
video_id (FK → VIDEOS.id)
position
created_at

Entity 10 — FAVORITES

id (PK)
user_id (FK → USERS.id)
content_type
content_id
created_at
UNIQUE(user_id, content_type, content_id)

Entity 11 — CATEGORIES

id (PK)
name
slug (UNIQUE)
description
kind
parent_id
created_at
updated_at

Entity 12 — BOOK_PLAYLISTS

id (PK)
title
description
thumbnail_url
created_by (FK → USERS.id)
is_public
created_at
updated_at
category_id (FK → CATEGORIES.id)

Entity 13 — BOOK_PLAYLIST_ITEMS

id (PK)

playlist_id (FK → BOOK_PLAYLISTS.id)
book_id (FK → BOOKS.id)
position
created_at

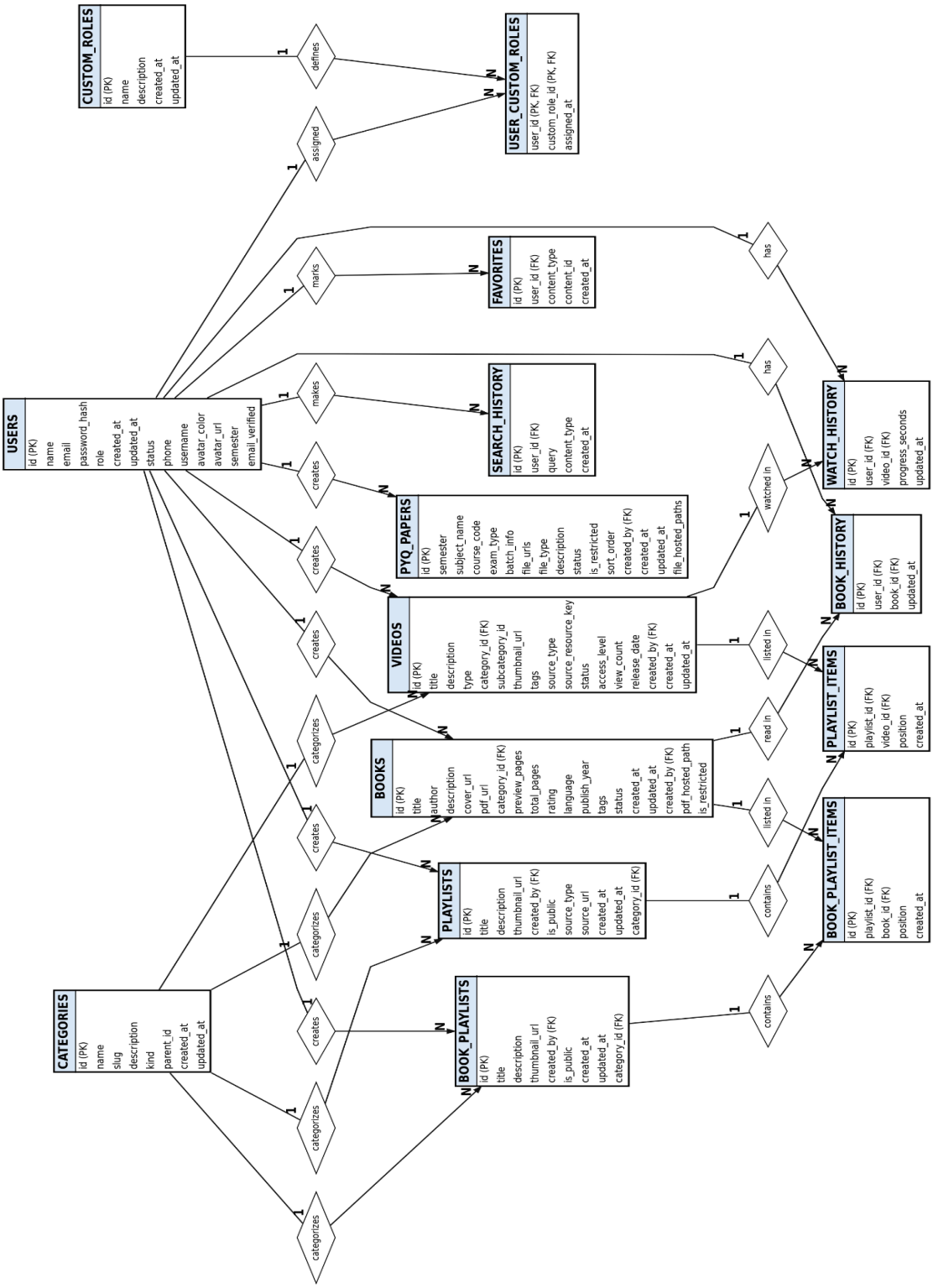
Entity 14 — BOOK_HISTORY

id (PK)
user_id (FK → USERS.id)
book_id (FK → BOOKS.id)
updated_at
UNIQUE(user_id, book_id)

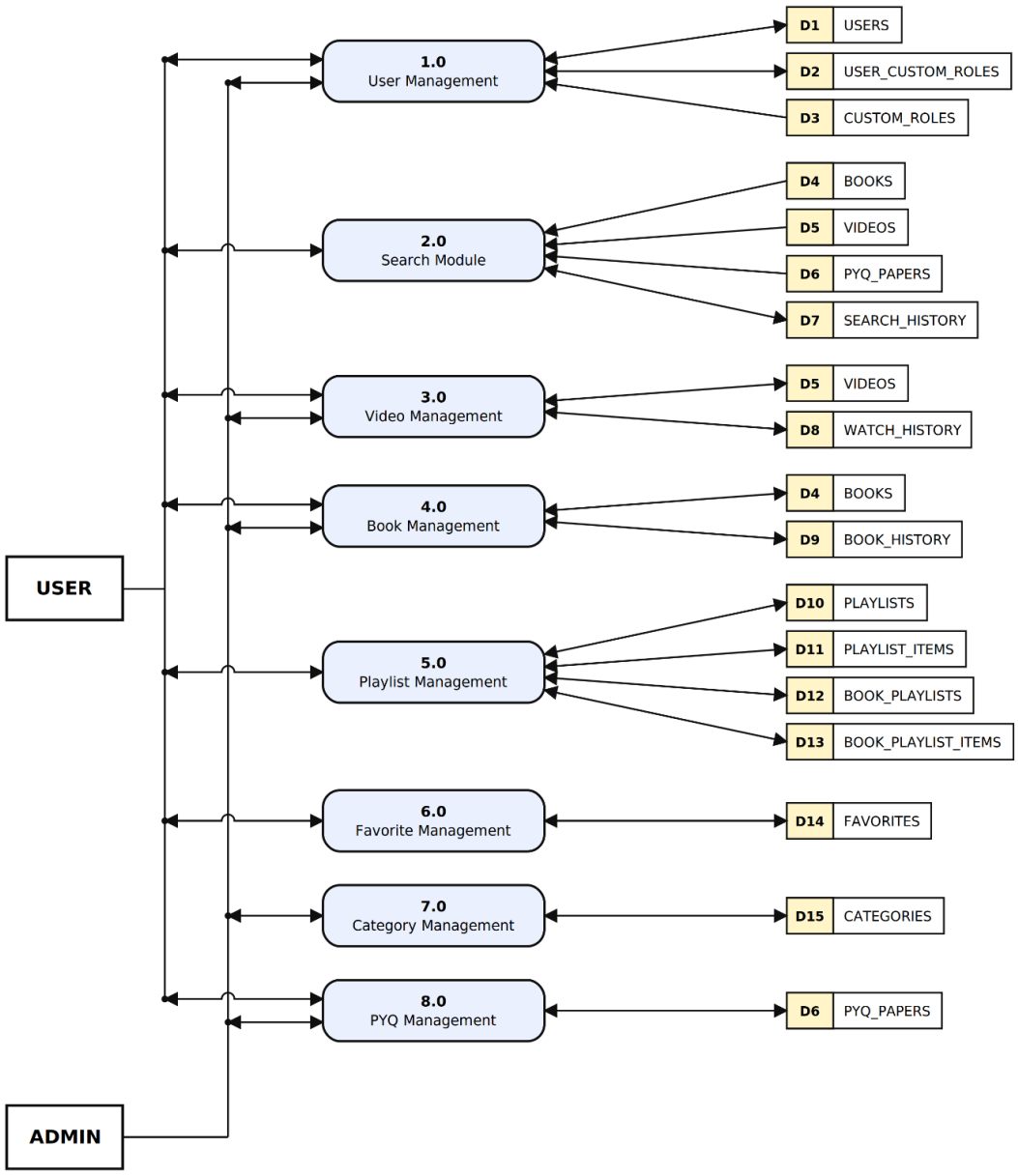
Entity 15 — CUSTOM_ROLES

id (PK)
name
description
created_at
updated_at

ER Diagram including Cardinality Specification:



Data Flow Diagram:



Schema Diagram:

